IASNLP 2026
Machine Learning

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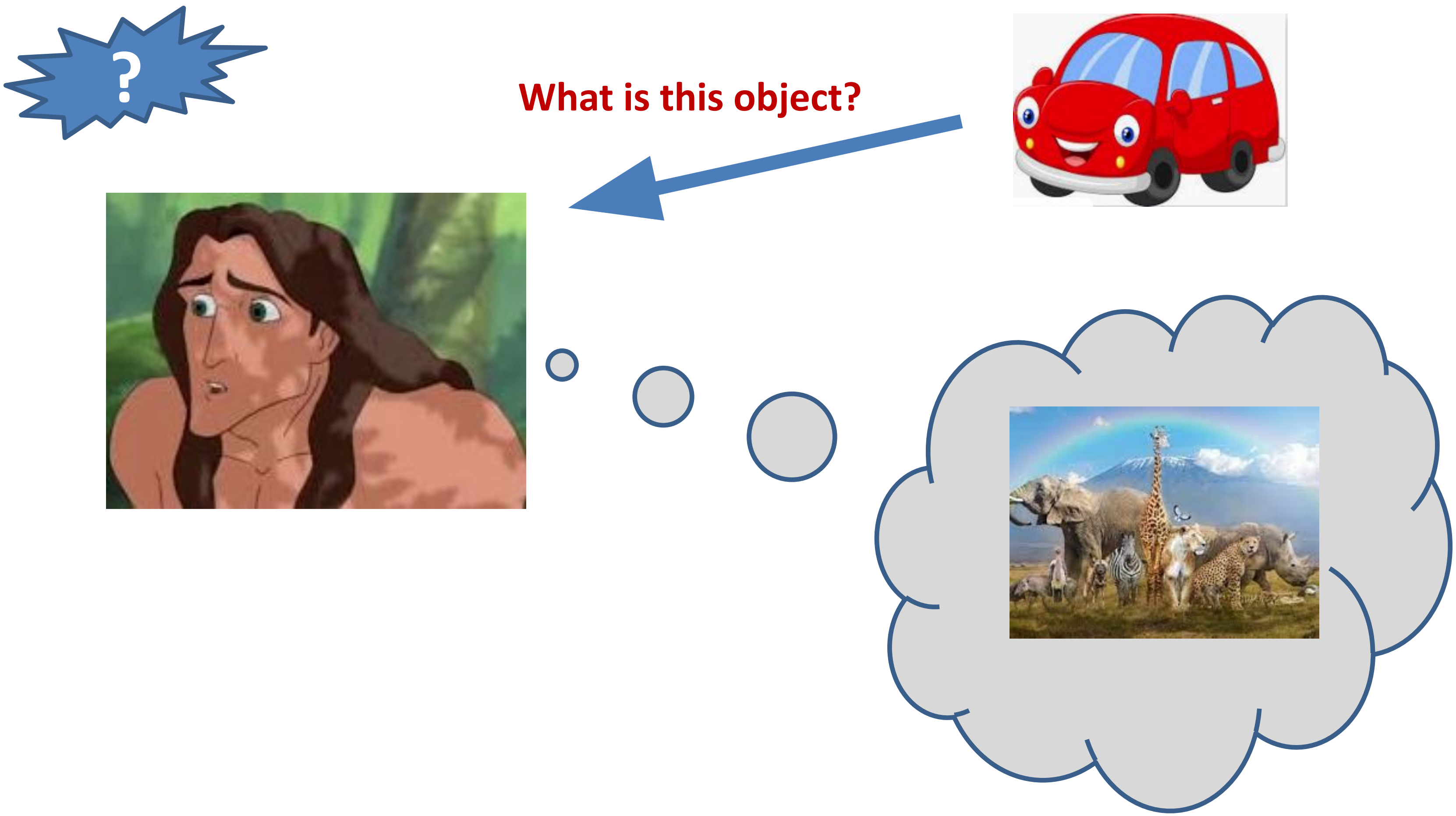
LTRC IIIT-H

Outline

1. Introduction
2. What is Machine Learning
3. Types of Machine Learning
4. Supervised Learning
    a. Regression
    b. Classification
5. Unsupervised Learning
    a. Clustering
    b. Association
6. Reinforcement Learning

Introduction

Introduction



Introduction

show him

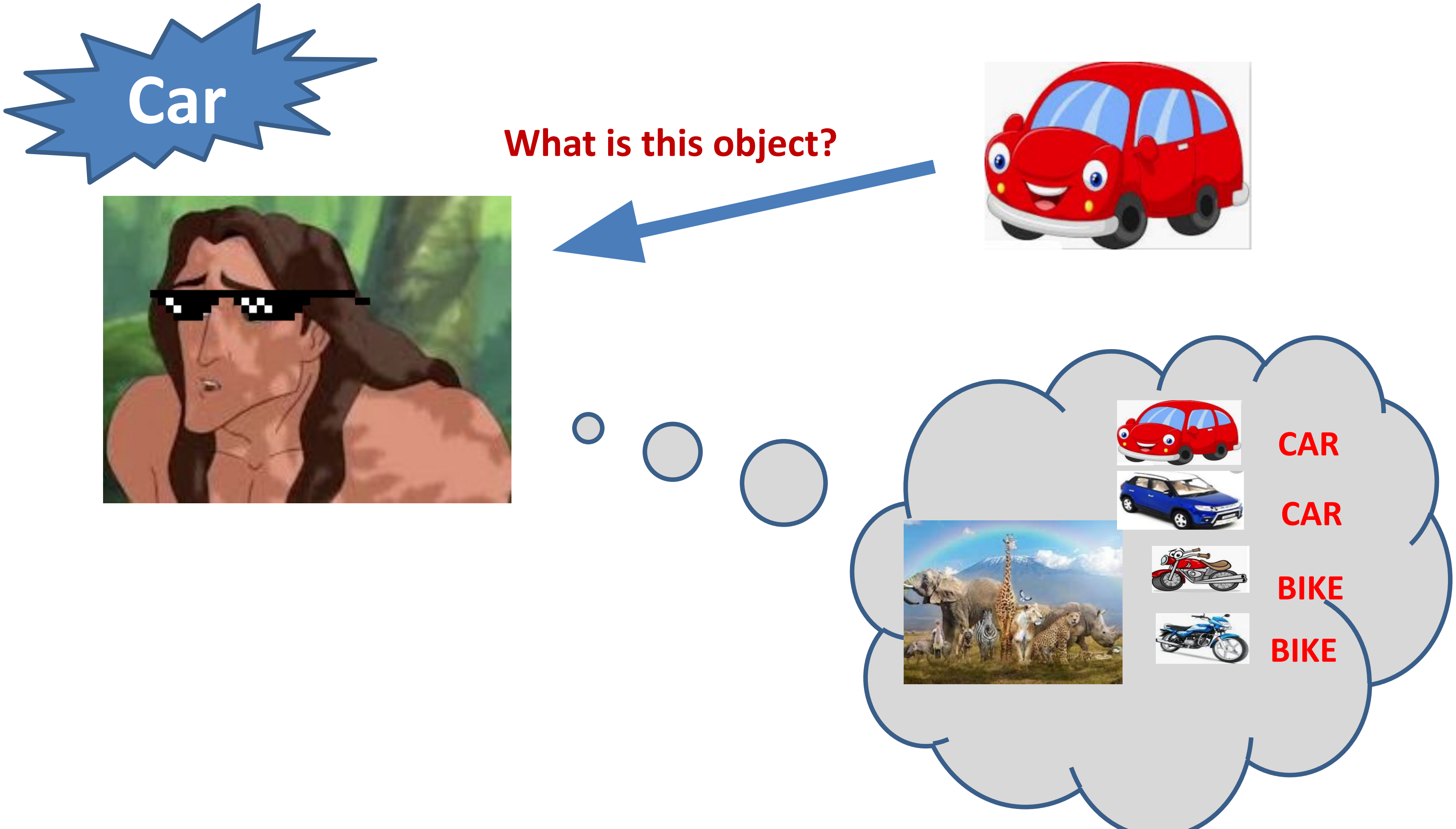
CAR

CAR

BIKE

BIKE

Introduction



What about a Machine?

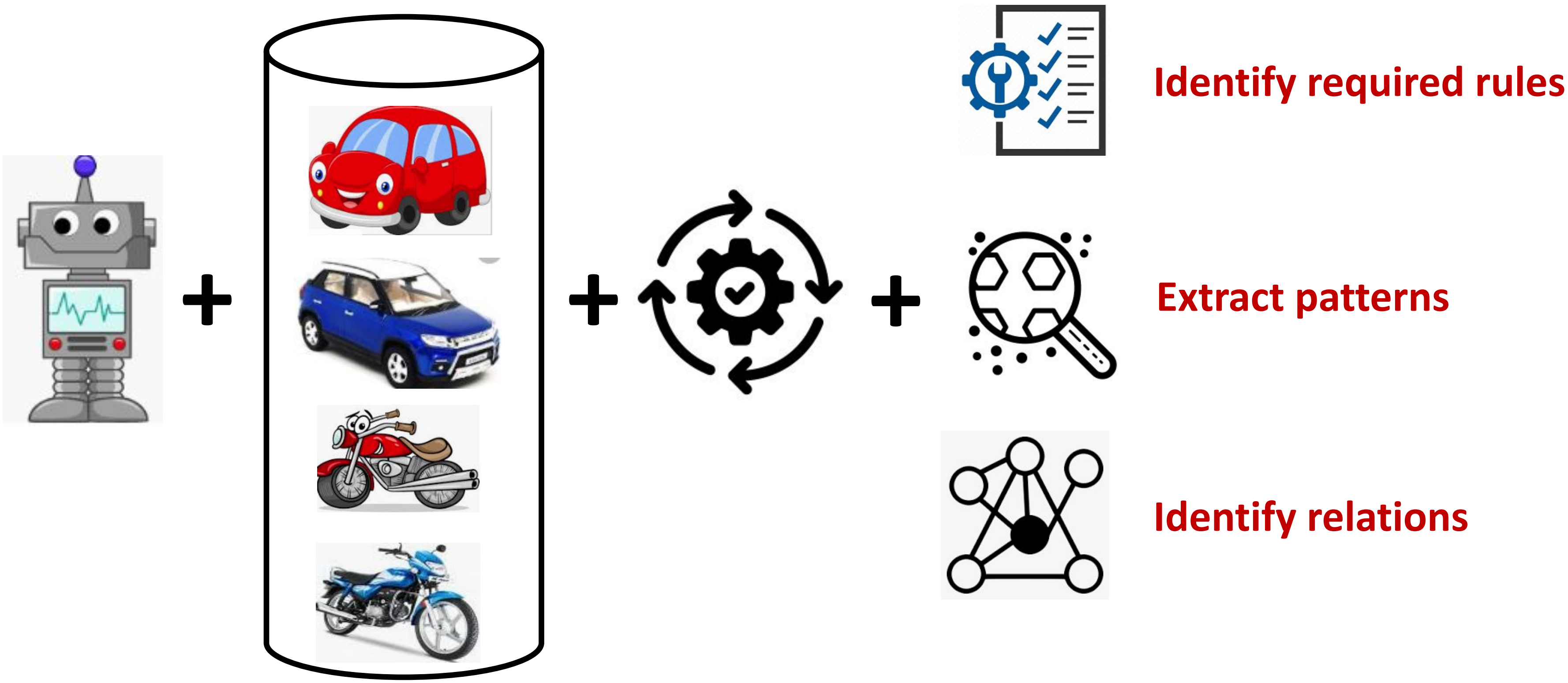
We can ask a machine to perform operations

- Arithmetic calculations (Addition, etc.)
- Face recognition and classification
- Natural Language Understanding

Machines follow instructions

What about a Machine?

The Machine Learning pipeline transforms raw datasets into actionable intelligence through a sequence of automated processes.



Dataset

1. Input Data

Collection of varied samples used for training.

2. Processing

Algorithms analyze data to identify features.

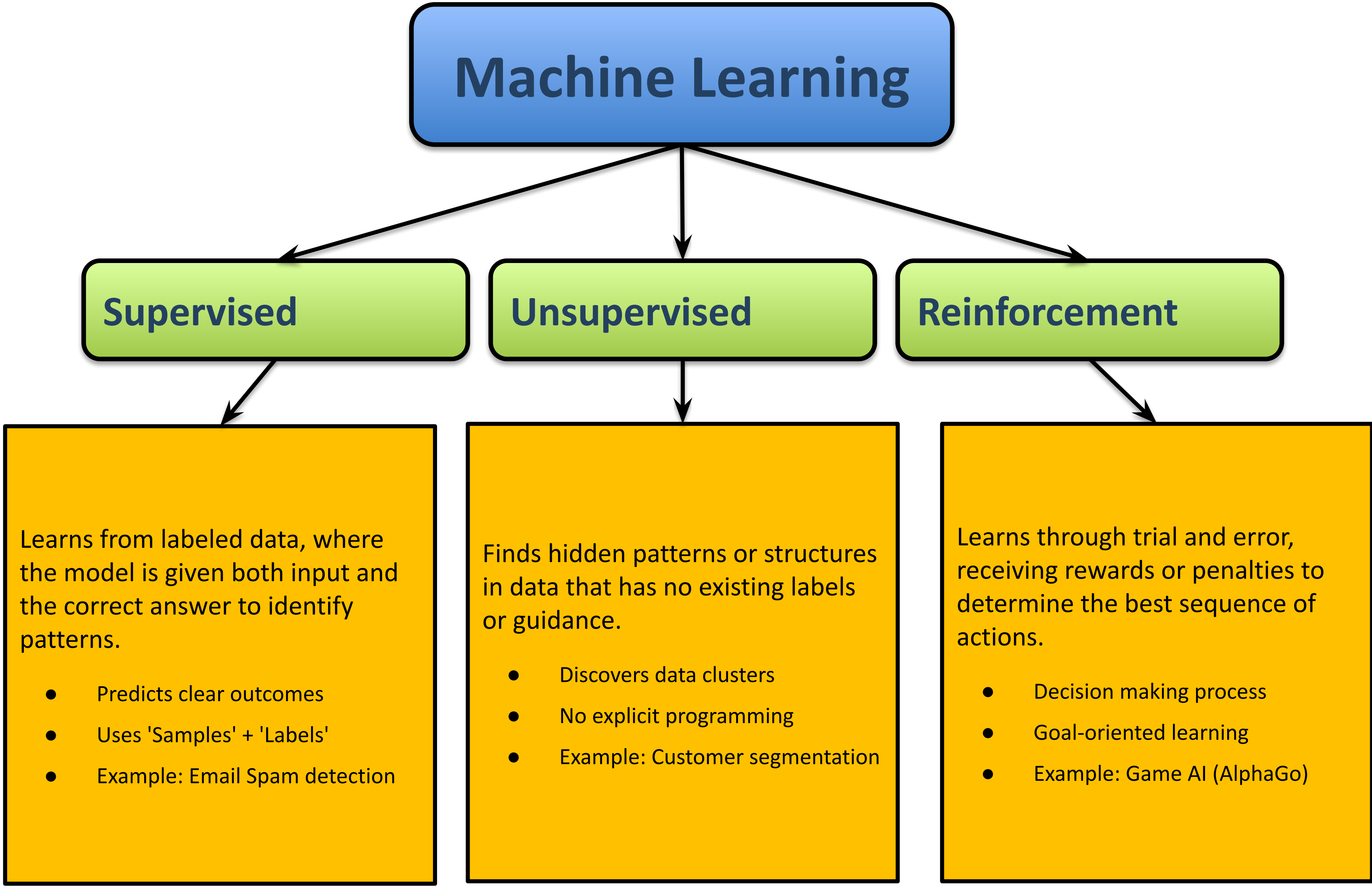
3. Outcome

Structural understanding through rules, patterns, and relations.

What is Machine Learning?

- **Machine learning is a branch of Artificial Intelligence that focuses on developing models and algorithms that let computers learn from data without being explicitly programmed for every task.**
- **In simple words, ML teaches the systems to think and understand like humans by learning from the data.**

Types of Machine Learning



Supervised Machine Learning

Samples (X)	Labels (Y)
	CAR
	CAR
	BIKE
	BIKE

Training Dataset

The model learns the mapping between samples and their corresponding labels to predict future data.

f( ) = CAR

Unsupervised Machine Learning

NO LABELS

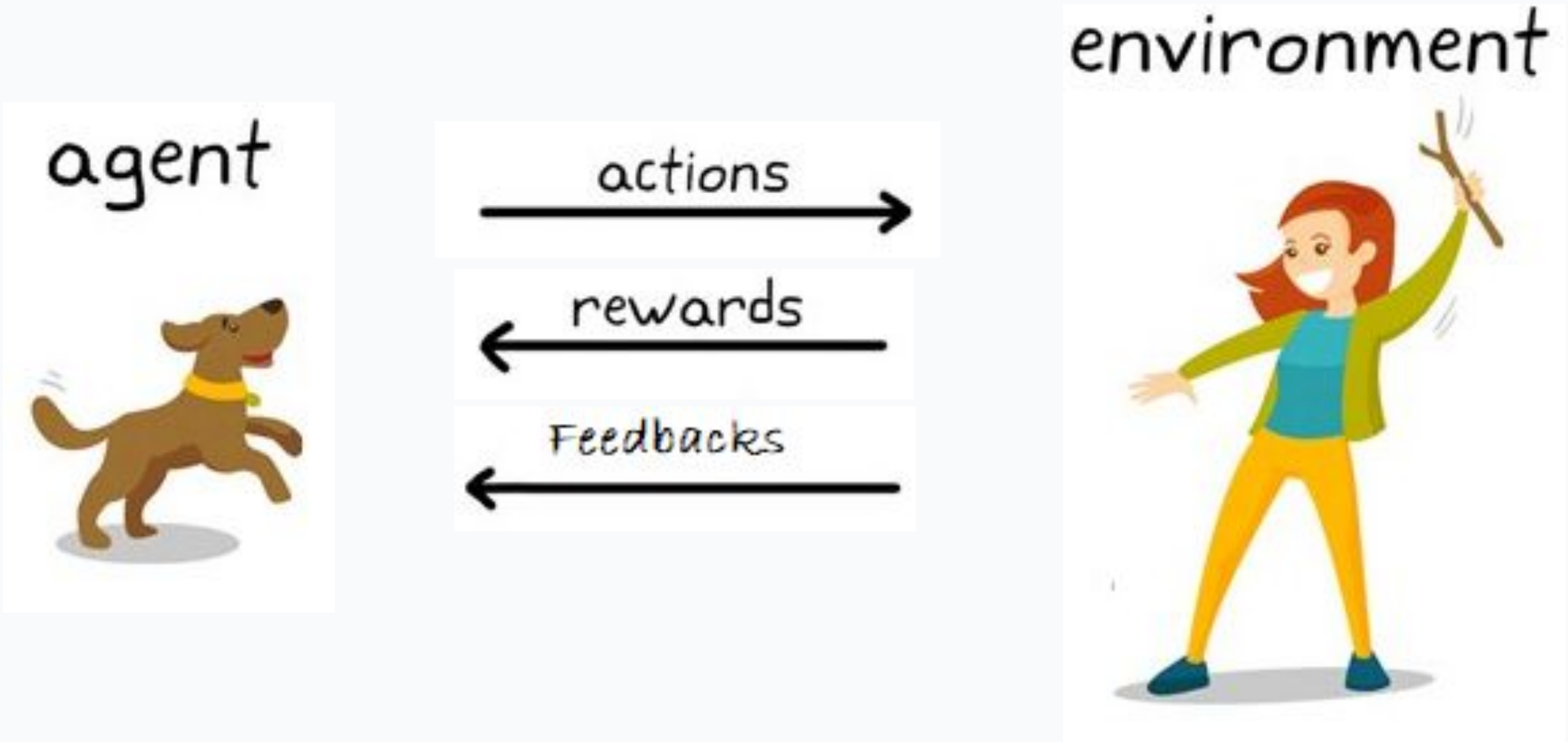
Samples (X)

Training Dataset

The model analyzes the samples without any labels to find hidden structures, clusters, or associations.

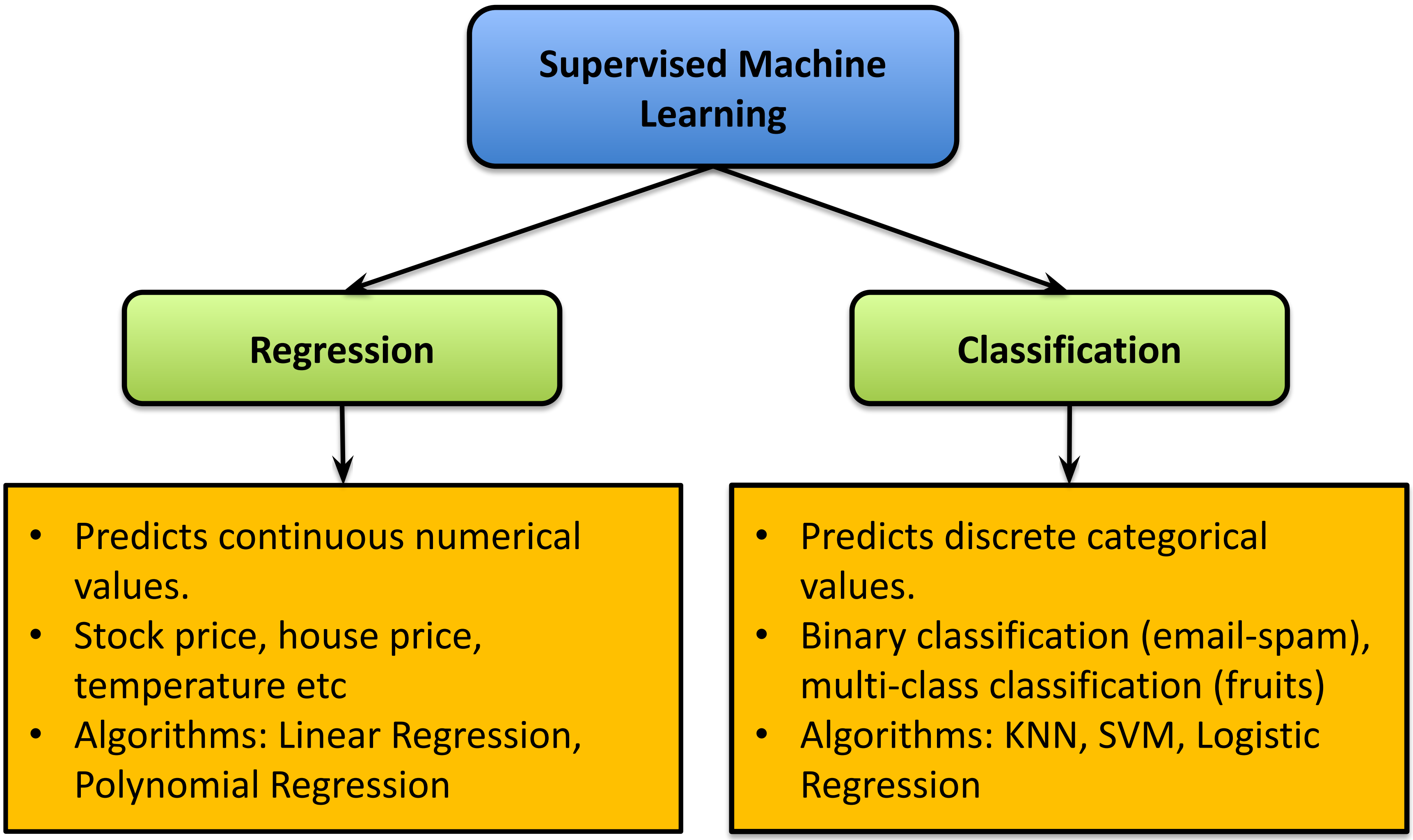
Goal: Discover Hidden Patterns

Reinforcement Machine Learning



Reinforcement Learning involves an **agent** taking **actions** in an **environment** to maximize cumulative **rewards** based on **feedback**.

Supervised Machine Learning



Linear Regression

- Regression algorithm
- A method used to model relationship between a dependent variable and one or more independent variables

Linear Regression & Loss Functions

Linear Regression Formula

$$y = \beta_0 + \beta_1 x + \varepsilon$$

- **Dependent Variable (y):** The outcome or target value we aim to predict.
- **Independent Variable (x):** The predictor or input feature used to explain variations in y.
- **β_0, β_1:** Coefficients representing the intercept and slope of the relationship.

Mean Squared Error (MSE) Loss

$$\mathrm{MSE} = (1/n)\ \Sigma(y_i - \hat{y}_i)^2$$

- **Actual Value (y_i):** The observed value of the dependent variable.
- **Predicted Value ($\hat{y}_i$):** The value calculated using the independent variables (x).
- **Squared Difference:** Penalizes larger errors more heavily to optimize the model fit.

Linear regression models the relationship between a **dependent variable** and one or more **independent variables**. The MSE loss function quantifies the average squared difference between predicted and actual values to refine these coefficients.

KNN (K-Nearest Neighbours)

- Classification algorithm
- Works by finding the "k" closest data points (neighbors) to a given input and makes a predictions based on the majority class
- The value of k in KNN decides how many neighbors the algorithm looks at when making a prediction.

KNN Distance Metrics & Selection

1. Euclidean Distance

$$d(x, y) = \sqrt{\Sigma}\, (x_i - y_i)^2$$

- **Criteria:** Best for continuous, well-scaled data where "as-the-crow-flies" distance matters.

2. Manhattan Distance

$$d(x, y) = \Sigma\, |x_i - y_i|$$

- **Criteria:** Preferred when data is high-dimensional or features are not well-scaled; less sensitive to outliers.

3. Minkowski Distance

$$d(x, y) = (\Sigma\, |x_i - y_i|\text{^}p)\text{^}(1/p)$$

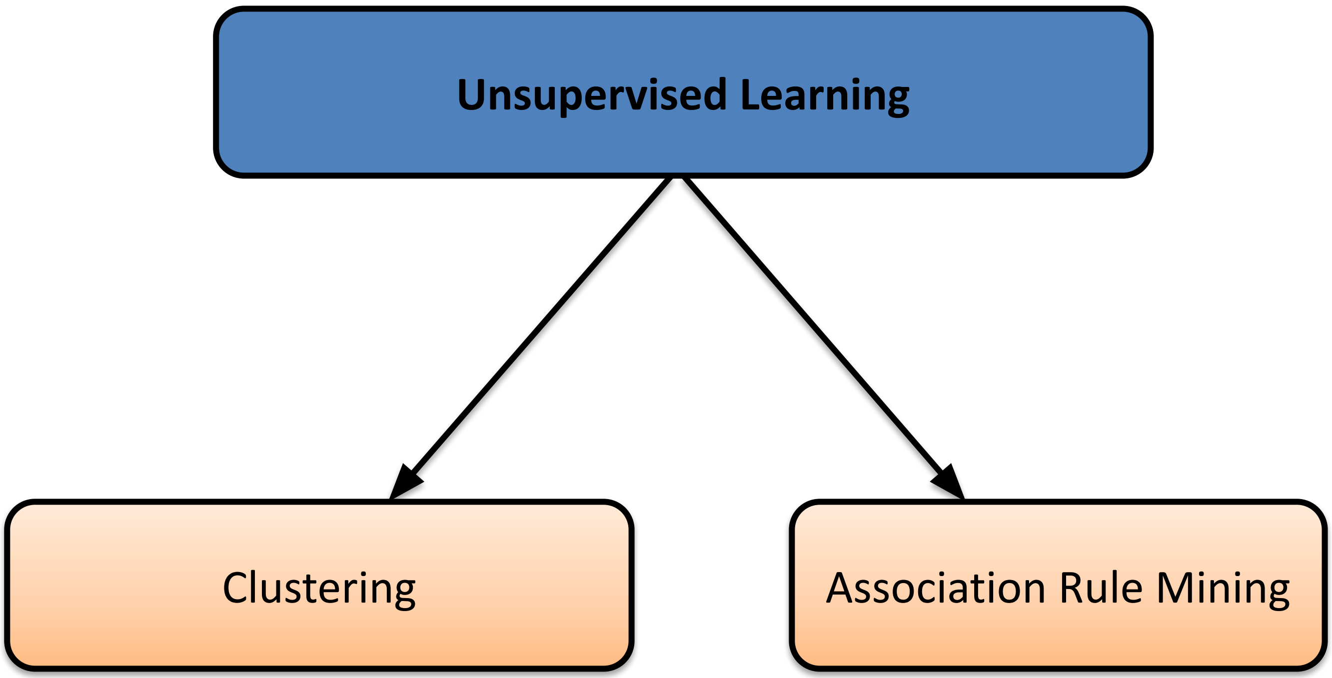
- **Criteria:** Generalized form. Tune 'p' (p=1 is Manhattan, p=2 is Euclidean) for maximum flexibility.

Unsupervised Learning

Data have no target attribute/ label

We have to find out some intrinsic structures/ patterns/ associations among them

Unsupervised Learning

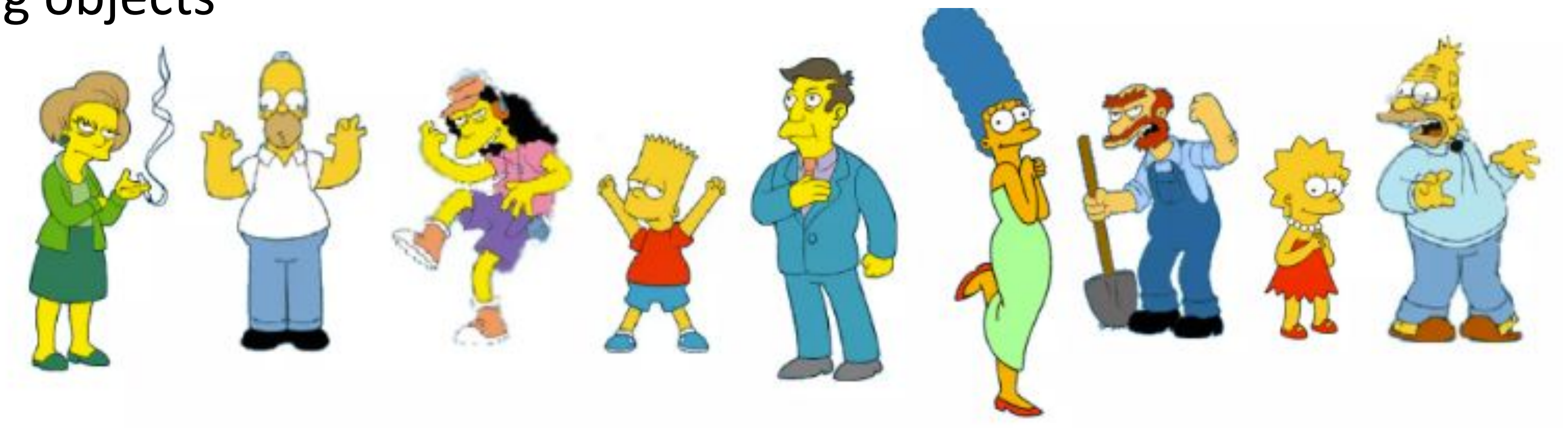


Clustering

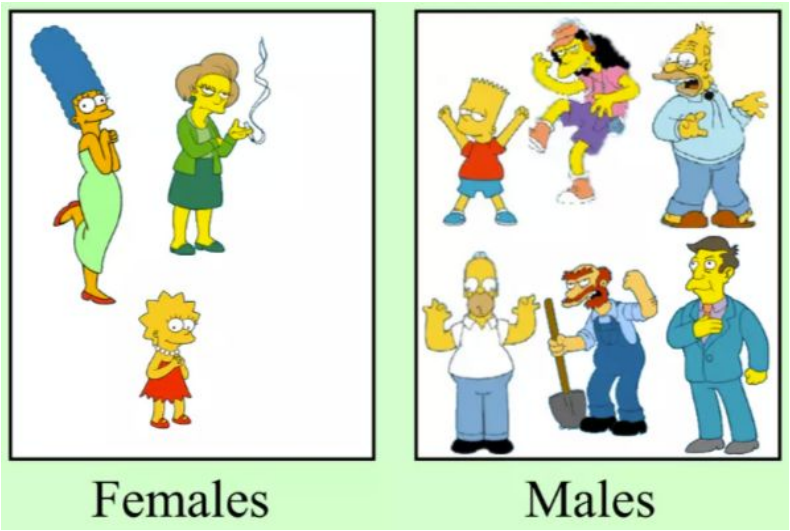
Organizing Data into clusters such that there is

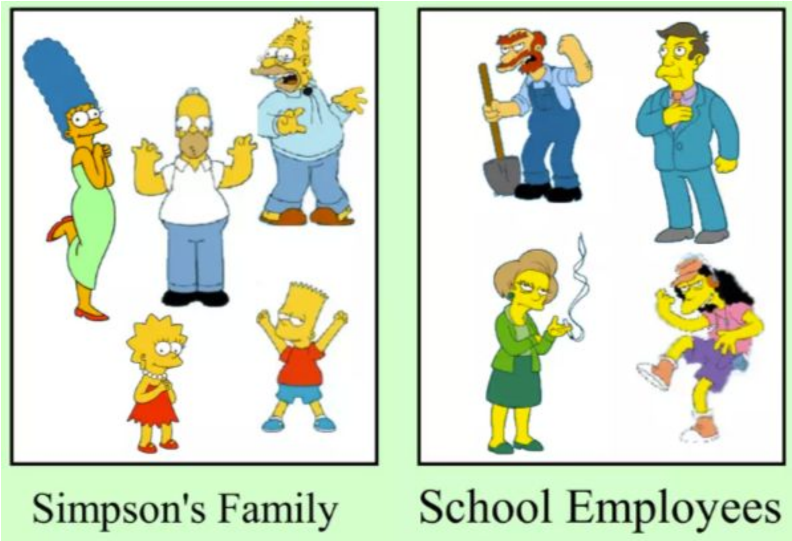
- High intra cluster similarity
- Low inter-cluster similarity

Informally- Finding natural groupings among objects



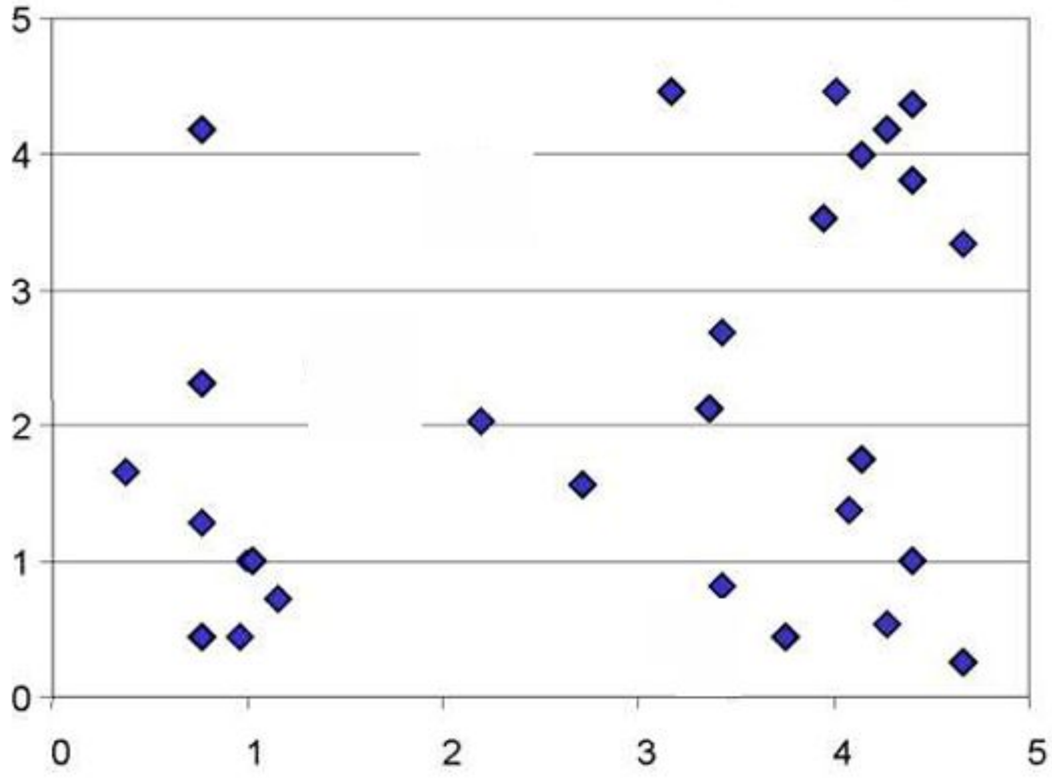
Clustering is Subjective





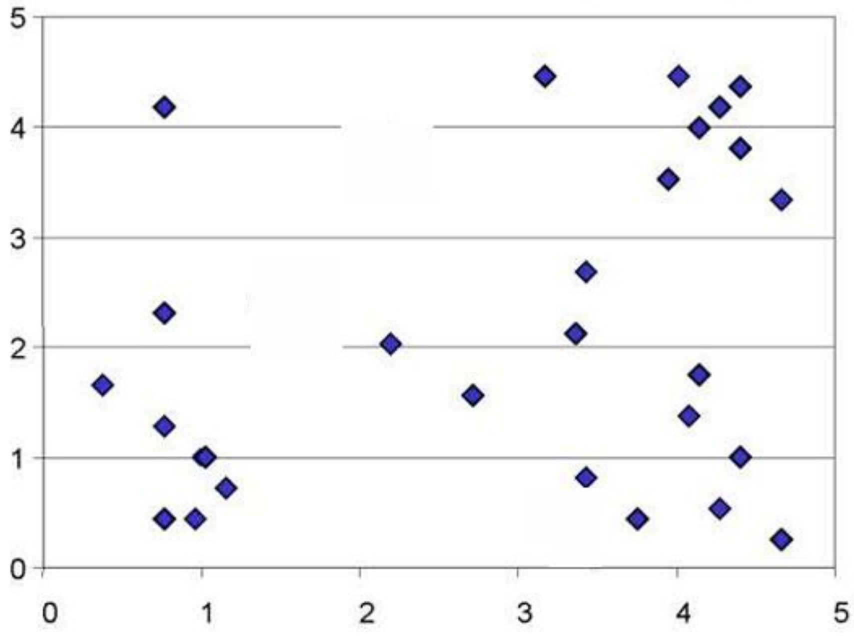
K Means Clustering Algorithm

Unlabelled data



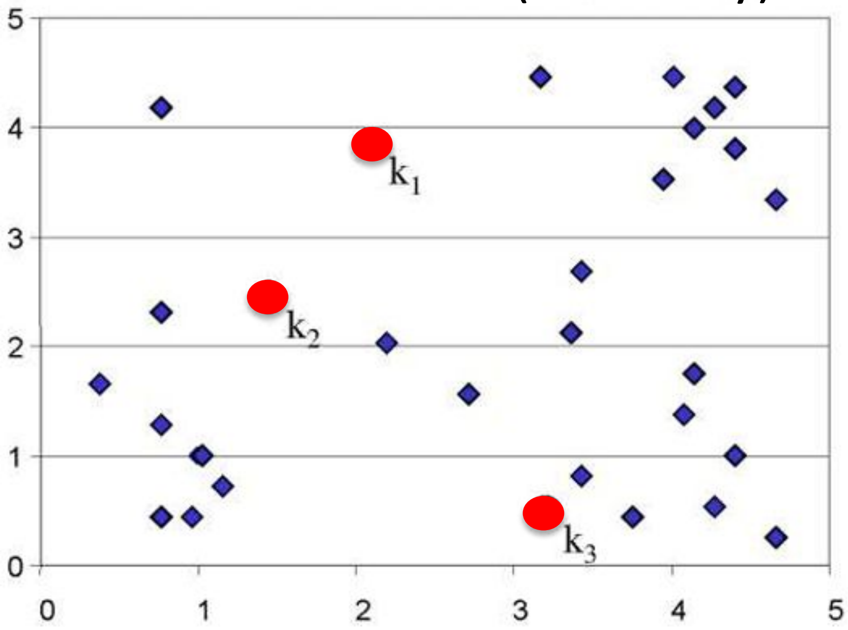
K Means Clustering Algorithm

Initially:



Decide k =3

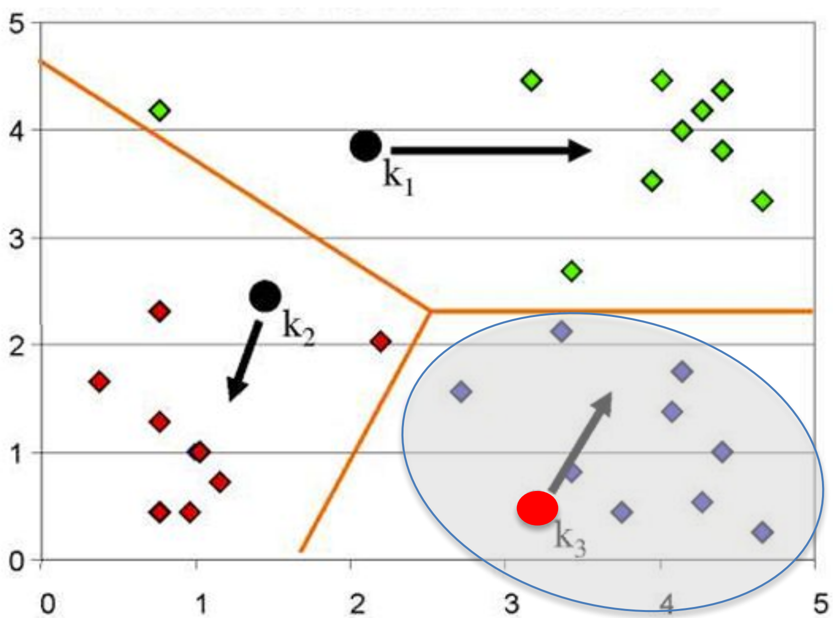
Initialize k centers (randomly)



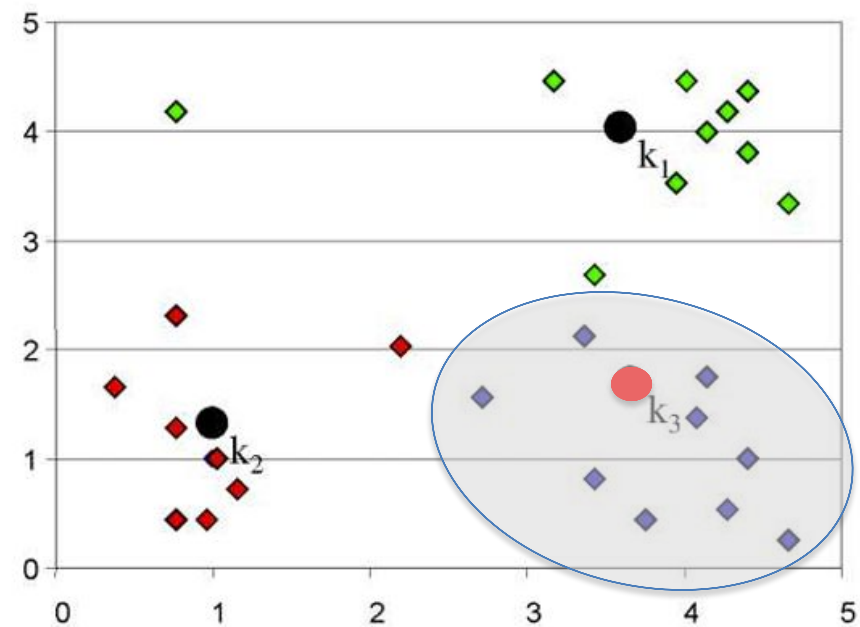
K Means Clustering Algorithm

Iteration 1:

Assign all objects to
nearest centers.



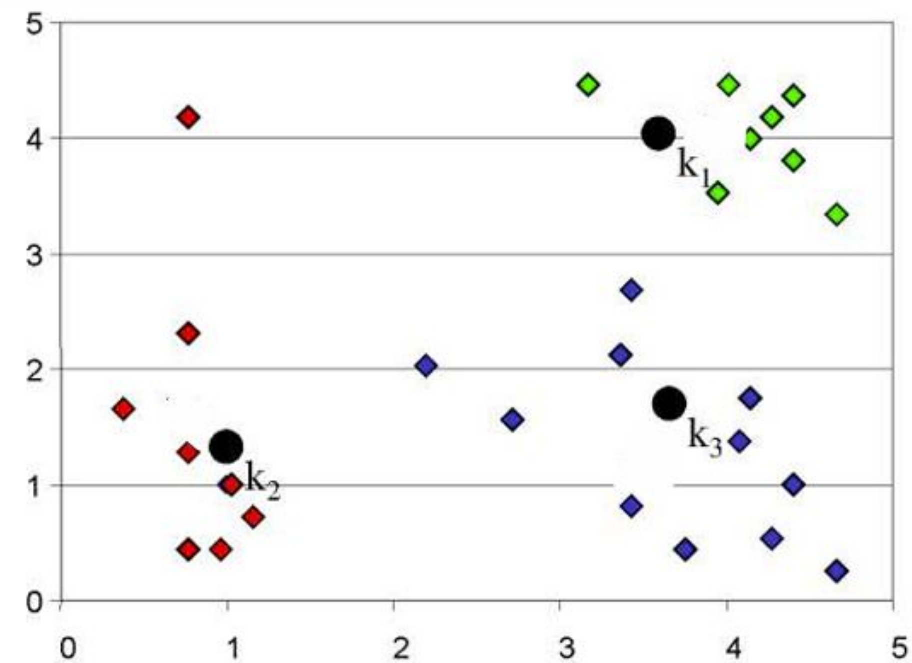
Move center to the mean
of its members



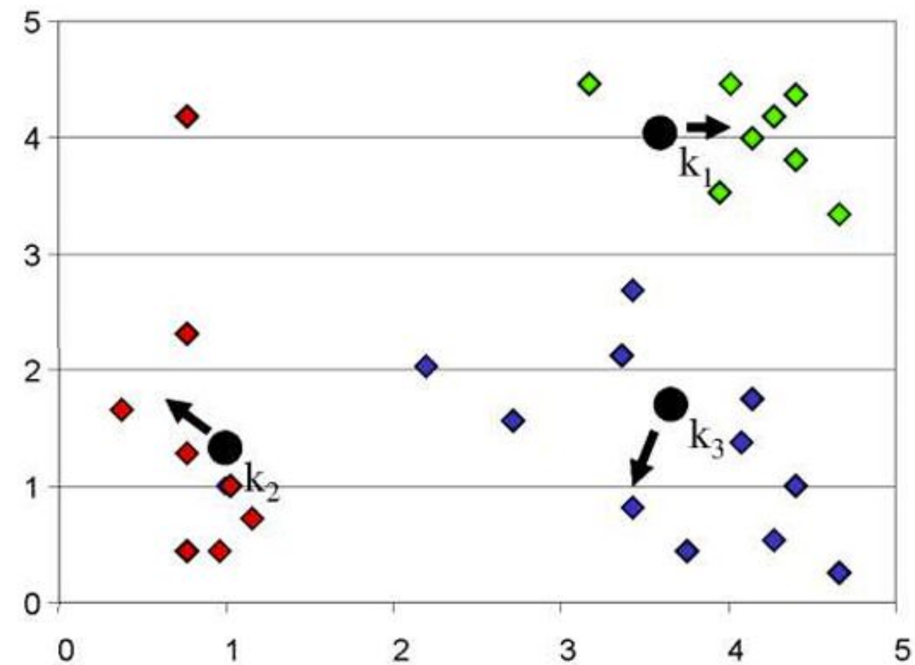
K Means Clustering Algorithm

Iteration 2:

Re-assign all objects to new centers.



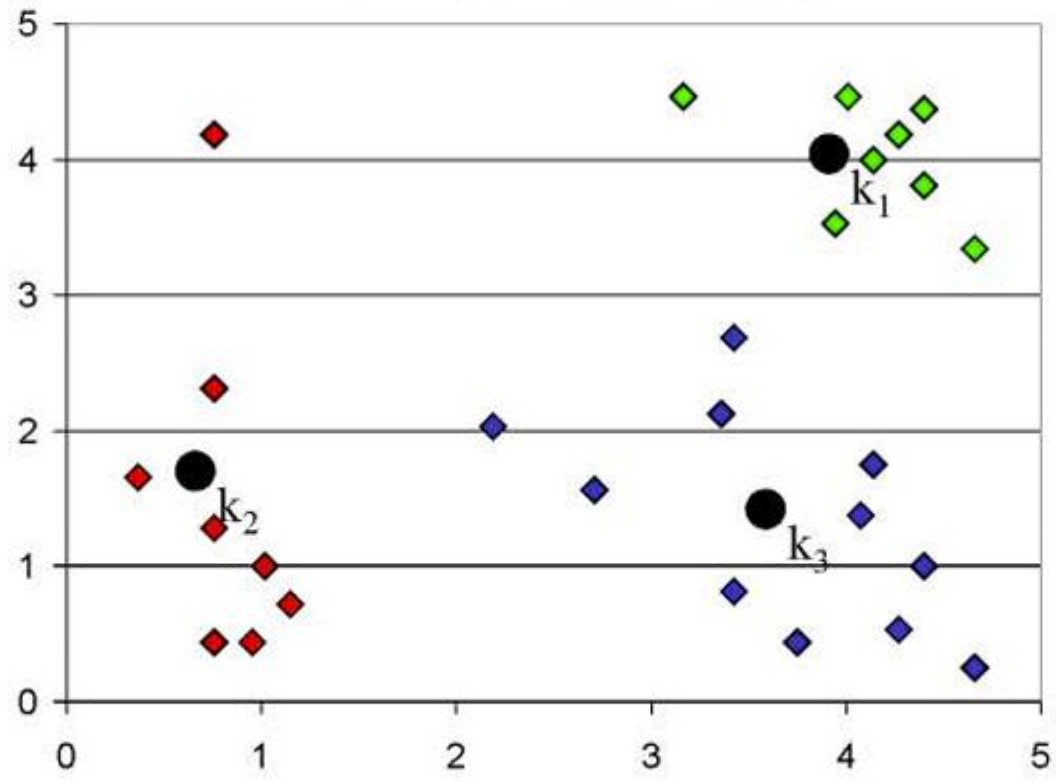
Move the centers agian



K Means Clustering Algorithm

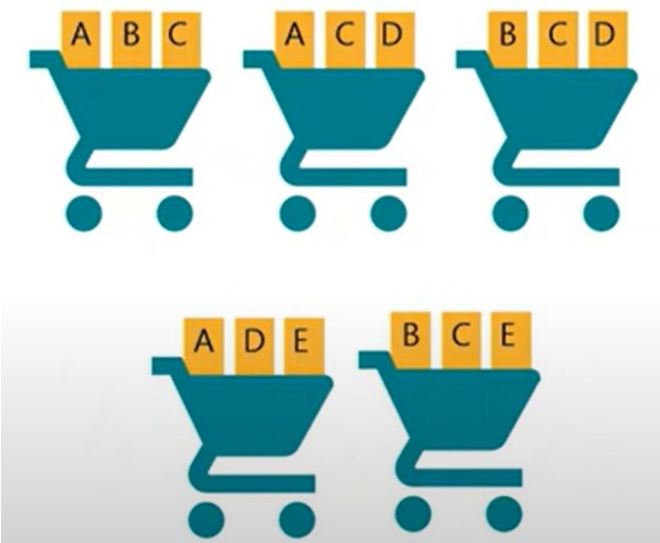
Finished:

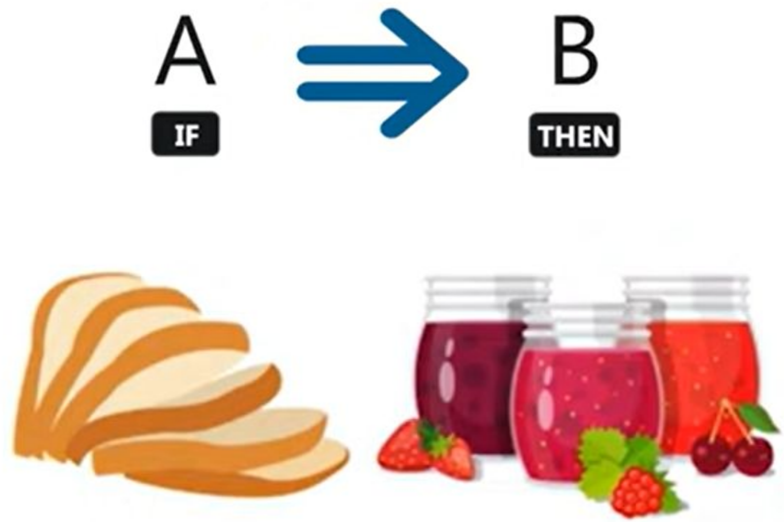
Re-assign and move centers, until…..
No objects membership is changed



Association Rule Mining

Unsupervised learning technique that checks the dependency of one data item on another data item and maps accordingly.





Market Basket Analysis

Apriori Algorithm

TID	List of Items IDs
T100	I1, I2, I5
T200	I2, I4
T300	I2, I3
T400	I1, I2, I4
T500	I1, I3
T600	I2, I3
T700	I1, I3
T800	I1, I2, I3, I5
T900	I1, I2, I3

1-item Sets	Frequency
I1	6
I2	7
I3	5
I4	4
I5	2

Minimum Support Count= 2

Apriori Algorithm

TID	List of Items IDs
T100	I1, I2, I5
T200	I2, I4
T300	I2, I3
T400	I1, I2, I4
T500	I1, I3
T600	I2, I3
T700	I1, I3
T800	I1, I2, I3, I5
T900	I1, I2, I3

2-item Sets	Frequency
I1, I2	4
I1, I3	4
I1, I4	1
I1, I5	2
I2, I3	3
I2, I4	2
I2, I5	2
I3, I4	0
I3, I5	1
I4, I5	0

Frequent 2-item Sets	Frequency
I1, I2	4
I1, I3	4
I1, I5	2
I2, I3	3
I2, I4	2
I2, I5	2

Minimum Support Count= 2

Apriori Algorithm

TID	List of Items IDs
T100	I1, I2, I5
T200	I2, I4
T300	I2, I3
T400	I1, I2, I4
T500	I1, I3
T600	I2, I3
T700	I1, I3
T800	I1, I2, I3, I5
T900	I1, I2, I3

3-item Sets	Frequency
I1, I2, I3	2
I1, I2, I4	0
I1, I2, I5	2
I1, I3, I4	1
I1, I3, I5	1
I1, I4, I5	0
I2, I3, I4	0
I2, I3, I5	1
I2, I4, I5	0
I3, I4, I5	0

Frequent 3-item Sets	Frequency
I1, I2, I3	2
I1, I2, I5	2

Minimum Support Count= 2

Apriori Algorithm

TID	List of Items IDs
T100	I1, I2, I5
T200	I2, I4
T300	I2, I3
T400	I1, I2, I4
T500	I1, I3
T600	I2, I3
T700	I1, I3
T800	I1, I2, I3, I5
T900	I1, I2, I3

4-item Sets	Frequency
I1, I2, I3, I5	1

Frequent 4-item Sets	Frequency
Not Possible	

Minimum Support Count= 2

Apriori Algorithm

Frequent
3-item sets-
{1, 2, 3}
{1, 2, 5}

{1, 2, 3}

{1} => {2,3}
{2} => {1,3}
{3} => {1,2}
{2,3} => {1}
{1,3} => {2}
{1,2} => {3]

Measure Association

$$\text{Support (A)} = \frac{\text{Number of transaction in which A appears}}{\text{Total number of transactions}}$$

$$\text{Confidence } (A \longrightarrow B) = \frac{\text{Support}(A \cup B)}{\text{Support}(A)}$$

Apriori Algorithm

{1} => {2,3}
{2} => {1,3}
{3} => {1,2}
{2,3} => {1}
{1,3} => {2}
{1,2} => {3]

Given:
Minimum Support = 2/9= 22.22%
Minimum Confidence = 50%

Rule 1: {1} → {2, 3} {S= 22.22 %, C=33.34%}

Support = 2/9 = 22.22%

Confidence = Support (1, 2, 3)/Support(1) = $\frac{2/9}{6/9}$ = 2/6 = 33.34% < 50%

Invalid Rule

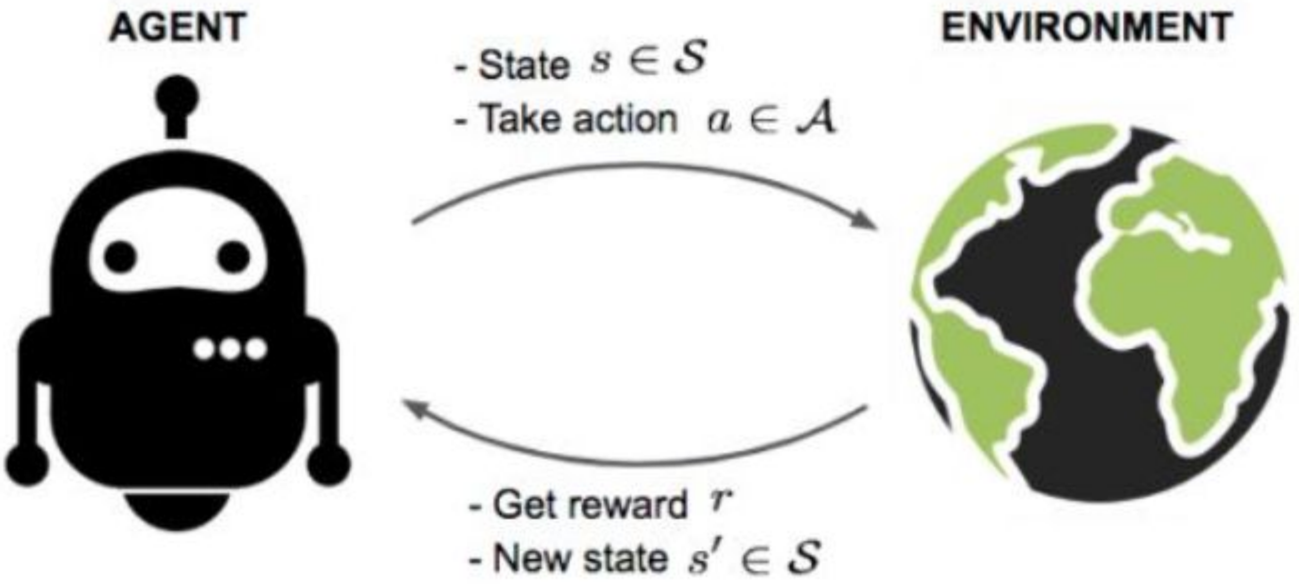
Rule 5: {1, 3} → {2} {S= 22.22 %, C=50%}

Support = 2/9 = 22.22 %

Confidence = Support (1, 2, 3)/Support(1, 3) = $\frac{2/9}{4/9}$ = 2/4 = 50% >= 50%

Valid Rule

Reinforcement Learning



Reinforcement Learning

Agent: the cat

State: the position of the cat (x, y) in the grid

Action: at each position, the cat can move to one of the 4-directionally connected cells. If a move is invalid, the cell will not move and remain in the same position. Every time the cat makes a move, it results in a new state and a reward.

Reward model:

- A move to another empty cell results in a reward of 0.
- A move towards the broom, will result in a reward of -1.
- A move towards the bathtub will result in a reward of -10 and the cat fainting (episode over). The cat will be respawned at the initial position again.
- A move towards the meat will result in a reward of +100

Policy: a policy rules how the agent selects the action to perform given the state it is in: $a_t \sim \pi(\cdot\,|s_t)$

The goal in RL is to select a policy that maximizes the expected return when the agent acts according to it.